

Calendly Marketing Insights

Data Model

1. Purpose

This document describes the proposed data model for the Calendly Marketing Insights pipeline. The model follows a Bronze, Silver, and Gold lakehouse structure, where raw source data is preserved in Bronze, cleaned analytical datasets are created in Silver, and dashboard-ready business metrics are produced in Gold. The data model is intended to support marketing campaign analysis, cost-per-booking calculations, booking trend analysis, channel attribution, time-slot analysis, and employee meeting load reporting.

2. Data Model Overview

The proposed data model follows a Bronze, Silver, and Gold lakehouse structure. The Bronze layer preserves raw source data, the Silver layer contains cleaned and normalized analytical datasets, and the Gold layer contains aggregated metric tables used by the Streamlit dashboard.

The primary analytical relationship in this project is between Calendly booking activity and marketing spend. Calendly booking data provides the lead and meeting activity, while marketing spend data provides the cost associated with each marketing channel. The two datasets will be associated primarily by marketing channel and date where applicable.

3. Proposed Table Inventory

“The final implementation may use S3 prefixes and Delta table names that follow these logical table names.”

Bronze:

- bronze_calendly_webhooks
- bronze_marketing_spend

Silver:

- silver_calendly_bookings
- silver_marketing_spend

Gold:

- gold_daily_calls_by_source
- gold_cpb_by_channel
- gold_booking_trends
- gold_channel_attribution
- gold_booking_volume_by_time_slot
- Gold_employee_meeting_load

4. Bronze Data Model

The Bronze layer will store raw source records with minimal transformation. These datasets are intended for preservation, traceability, and reload support rather than direct reporting.

Bronze Calendly Webhooks

This dataset will store raw Calendly webhook events as they are received from the API Gateway and Lambda ingestion process.

Grain: one raw webhook event per record.

Primary fields include:

- Webhook event type
- Webhook created timestamp
- Raw Calendly payload
- Source system
- Ingestion timestamp
- Raw S3 object path
- Processing date

Bronze Marketing Spend

This dataset will store raw marketing spend JSON records or raw spend files as ingested from the public S3 source.

Grain: one raw spend record per source file, with source file metadata preserved.

Primary fields include:

- Spend date
- Marketing channel
- Spend amount

- Source file name
- Source file path
- Ingestion timestamp
- Processing date
- Raw payload

5. Silver Data Model

The Silver layer will contain cleaned, validated, flattened, and typed datasets. These tables provide the reusable analytical foundation for Gold metrics.

Silver Calendly Bookings

This table will contain one row per Calendly invitee booking after flattening the nested webhook JSON structure.

Grain: one booking/invitee per row.

Primary key: booking_id.

Primary fields include:

- Booking ID
- Invitee URI
- Scheduled event ID
- Scheduled event URI
- Calendly event type URI
- Mapped marketing channel
- Invitee email
- Invitee name
- Invitee timezone
- Invitee status
- Rescheduled flag
- Meeting name
- Meeting status
- Meeting start timestamp
- Meeting end timestamp
- Booking date
- Meeting date
- Meeting hour
- Meeting day of week
- Meeting week start date
- UTM source
- UTM medium

- UTM campaign
- UTM content
- UTM term
- Salesforce UUID
- Employee ID
- Employee email
- Employee name
- Ingestion timestamp
- Source S3 path

The Calendly event type URI will be mapped to one of the supported marketing channels: Facebook paid ads, YouTube paid ads, or TikTok paid ads.

Silver Marketing Spend

This table will contain standardized marketing spend records.

Grain: one spend date and channel per row.

Primary key: spend_date plus channel.

Primary fields include:

- Spend date
- Marketing channel
- Spend amount in USD
- Source file name
- Source file path
- Ingestion timestamp
- Processing date

The Silver marketing spend table will validate that each record contains a supported channel and a valid numeric spend amount.

6. Gold Data Model

The Gold layer will contain business-ready metric tables for dashboard reporting. These tables will be built from the Silver Calendly bookings and Silver marketing spend datasets.

Gold Daily Calls by Source

Grain: booking date and channel.

Primary fields include:

- Booking date
- Channel
- Booked calls

This table supports the daily calls booked by source dashboard view.

Gold Cost Per Booking by Channel

Grain: channel, with optional date range filtering in the dashboard.

Primary fields include:

- Channel
- Total spend
- Total bookings
- Cost per booking

Cost per booking is calculated as total spend divided by total booked calls.

Gold Booking Trends

Grain: booking date and channel.

Primary fields include:

- Booking date
- Channel
- Booked calls
- Cumulative bookings, if needed for the dashboard

This table supports daily or weekly trend analysis.

Gold Channel Attribution

Grain: channel and campaign, where campaign data is available.

Primary fields include:

- Channel
- Campaign
- Total bookings
- Total spend
- Cost per booking

This table supports attribution analysis using mapped event types and UTM campaign fields.

Gold Booking Volume by Time Slot

Grain: meeting day of week, meeting hour, and channel.

Primary fields include:

- Meeting day of week
- Meeting hour
- Channel
- Booking count

This table supports analysis of when leads prefer to schedule calls.

Gold Employee Meeting Load

Grain: employee and week.

Primary fields include:

- Employee ID
- Employee email
- Employee name
- Week start date
- Weekly meeting count

The dashboard may also calculate or display average meetings per week across the selected reporting period.

7. Join Logic

Calendly booking records and marketing spend records will be joined primarily by channel and date when calculating channel-level performance metrics.

The Calendly booking channel will be derived from the Calendly event type URI. UTM fields will be used where available to support more detailed attribution. Marketing spend will be grouped by spend date and channel.

The primary relationship can be summarized as:

- Silver Calendly Bookings: booking_date, channel, booking_id
- Silver Marketing Spend: spend_date, channel, spend_usd

Join condition for date-level spend metrics:

- Calendly booking_date = Marketing spend_date
- Calendly channel = Marketing spend channel

For broader channel-level metrics, spend and bookings may also be aggregated by channel across the selected reporting period.

8. Data Model Summary

The Silver Calendly bookings table provides booking, meeting, channel, UTM, and employee-level detail.

The Silver marketing spend table provides date and channel-level spend detail.

The Gold tables aggregate those Silver datasets into business metrics for reporting and visualization.

This model supports the required dashboard views while preserving enough detail in Silver to rebuild or extend Gold metrics as business needs evolve.